

Warehouses and Diplomas:

Amazon's Expansion and Local Educational Attainment

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Abstract

Booms in low-skill work typically reduce schooling. I study the rollout of Amazon's logistics facilities, which entered 188 U.S. commuting zones between 2010 and 2024. I exploit variation in the timing of Amazon's entry across commuting zones, which brought both warehouse jobs and skilled jobs to local labor markets. I find that Amazon's arrival raised the share of adults with a bachelor's degree by 0.48 percentage points and the share with a master's degree by 0.39 points, with effects concentrated in college and graduate degrees and growing over time. The gains appear to come through two margins. Graduates stay in or move to these areas, and students switch from part-time to full-time study. These effects appear to reflect the skilled jobs that arrive with Amazon, which made a college degree more useful locally without raising the pay of college graduates.

Keywords: Amazon; e-commerce; educational attainment; college enrollment; human capital; local labor markets; difference-in-differences

JEL codes: I24; I25; J24; L81; R23

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1 Introduction

When a new employer arrives in a local labor market and offers jobs that do not require a degree, young people face a simple trade-off: stay in school, or take the job. A large literature finds that many of them take the job. Coal, fracking, and housing booms raised the wages of low-skill work and pulled teenagers and young adults out of school ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [Charles et al., 2018](#)), and in Mexico the expansion of export manufacturing raised school dropout among young people who were about 16 when the new factory jobs arrived ([Atkin, 2016](#)). Over the past fifteen years, the United States has experienced a national boom in low-skill work. Amazon expanded from a small number of U.S. fulfillment centers to more than one thousand facilities, entering local labor markets across the country with the support of billions of dollars in state and local subsidies ([Good Jobs First, 2022](#)). Its warehouses hire large numbers of workers without a college degree, at wages above the local median for such work ([Jones and Zipperer, 2018](#); [Cunningham, 2024](#)). The boom literature therefore predicts that Amazon’s arrival should lower local educational attainment.

There is a reason to expect the opposite. An Amazon facility is a warehouse, but it is also a node of one of the most technology-intensive logistics systems in the world. Its arrival brings engineers, managers, and technical staff, and it can draw suppliers and business services that hire college graduates ([Greenstone et al., 2010](#); [Rossi-Hansberg et al., 2021](#)). If this second layer of demand is large enough, Amazon’s entry raises the value of a degree in the local labor market rather than lowering it. Local governments compete for Amazon’s facilities with tax incentives, and the case for these incentives rests partly on their long-run effects on local workers ([Bartik, 2020](#)). Whether Amazon’s arrival raises or lowers local education matters for the subsidies that states and cities offer to attract it. Existing studies find that Amazon’s entry raises local employment and income ([Pathania and Netessine, 2025](#); [Cunningham, 2024](#)), but not whether it changes the education of the local population.

In this paper, I estimate the effect of Amazon’s entry on local educational attainment and college enrollment. I combine the opening date and location of every Amazon facility, from MWPVL International ([MWPVL International, 2025](#)), with a panel of 297 U.S. commuting zones built from published American Community Survey estimates and IPUMS microdata for 2005–2024, college enrollment from the Integrated Postsecondary Education Data System, and employment and establishment counts from County Business Patterns and the Business Dynamics Statistics. Amazon entered 188 of these commuting zones between 2010 and 2024. I compare commuting zones Amazon has already entered with those it will enter later, using the staggered difference-in-differences estimator of [Callaway and Sant’Anna \(2021\)](#). The main threat to this design is that Amazon chooses where and when to enter: it went first to the largest and richest markets, which are also more

educated. I show that although early and later entrants differ in levels, the shares of adults with a bachelor's or a master's degree were growing at the same rate in both groups before Amazon arrived.

I find that Amazon's entry raises local educational attainment. The share of adults aged 25 and older with a bachelor's degree rises by 0.48 percentage points, about 2 percent of its pre-entry level, and the effect grows over time, reaching 1.3 percentage points six years after entry. The share with a master's degree rises by 0.39 percentage points, reaching about 0.8 point after six years. The shares with a high school diploma or some college do not rise. These gains could reflect either more residents earning degrees or more degree holders moving in. The evidence points to both, with residents accounting for most of the increase. The number of adults with a bachelor's degree rises by 2.3 percent, about twice as fast as the number without one. Place of birth, which does not change over a lifetime, shows that graduates born in another state grow no faster than those born in their state of residence, while foreign-born graduates grow faster, by 6 percent. Among people who moved from another state or from abroad in the past year, the number with a bachelor's degree rises by about 7 percent and the number with a master's degree by about 11 percent, while the number without a degree does not change. Amazon's entry therefore attracts educated movers. This inflow is modest, however: even if every additional graduate who moved in stayed, it would account for less than half of the increase in the bachelor's share. The rest comes from people already living in these areas, either residents who earn degrees or graduates who stay rather than leave.

College enrollment shows how local residents respond. The share of college students who study part time falls by 1.2 percentage points, as full-time enrollment grows at public four-year colleges and part-time enrollment falls at public two-year colleges. Because full-time students tend to finish degrees faster, this shift might raise attainment within a few years. First-time enrollment at local four-year colleges, measured following [Charles et al. \(2018\)](#) and excluding colleges that draw most of their students from other states, rises by 0.22 percentage points of the population aged 18–24, about 4 percent of its pre-entry level, though this estimate is less precise than the attainment results, and the increase is concentrated among full-time students. Amazon's entry therefore does not appear to pull young people out of school; it appears to shift them toward full-time study at four-year colleges.

I show that the main finding is robust to a range of additional checks: alternative inference (randomly reassigning Amazon's entry years across commuting zones, where at most one of 500 random assignments produces an effect as large as the actual one, randomization-inference $p \leq 0.002$), falsification tests (moving Amazon's entry five years earlier, which produces small and insignificant placebo effects), alternative samples (ending the sample in 2019, before the pandemic), and an alternative estimator (the synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#), which gives effects of 0.67 percentage points on the bachelor's share and 0.45 on the master's share). The inference and pandemic checks use the microdata panel. Summarizing, Amazon's entry raises the share of adults with a bachelor's or a master's degree, and this increase is robust to

alternative samples, data, and estimators.

Why does Amazon’s entry raise education when other booms lowered it? I examine two channels. The first is income. Household income rises modestly, by 1.4 percent, while the poverty rate does not change, while rents rise faster, and the income and education gains are concentrated in commuting zones that were already better off rather than in the poorer places where financial constraints bind most. The income channel is therefore unlikely to explain the rise in attainment. The second channel, the demand for skilled work, fits the evidence closely. After Amazon arrives, professional, scientific, and technical establishments grow by 3.3 percent, and local employment shifts toward STEM occupations, especially computer and mathematical jobs, whose share rises by about 10 percent, and toward managerial and abstract-task work. The share of jobs held by college graduates rises by 0.6 percentage points. The college wage premium does not change. Amazon’s entry increases the number of skilled jobs close to home rather than their pay, and the evidence suggests that residents respond by investing in a college education and that graduates stay in the area.

This paper contributes to three literatures. First, it adds to the study of how local labor demand shocks shape educational investment ([Black et al., 2005](#); [Cascio and Narayan, 2022](#); [Kearney and Wilson, 2018](#); [James, 2017](#); [Charles et al., 2018](#); [Atkin, 2016](#)). These studies document that booms in low-skill work reduce schooling. I show that a boom led by a single large employer can do the opposite when it expands skilled work alongside low-skill work, so that the skill content of the shock, not only its size, determines its effect on education. Second, it contributes to the evaluation of place-based policies and of the incentives local governments offer large employers ([Busso et al., 2013](#); [Greenstone et al., 2010](#); [Bartik, 2020](#)), by documenting a human-capital return that such evaluations usually leave out. Third, it adds to the recent work on Amazon’s local effects ([Pathania and Netessine, 2025](#); [Cunningham, 2024](#)) and on how workers respond to local demand shocks through migration ([Notowidigdo, 2020](#); [Monras, 2018](#)), by showing that Amazon’s entry changes both who earns degrees and where degree holders choose to live.

The rest of the paper is organized as follows. Section 2.1 describes Amazon’s distribution network and presents a conceptual framework. Section 3.1 describes the data and Section 3.2 the empirical strategy. Section 4 presents the effects on attainment and enrollment, Section 5 examines mechanisms, Section 6 reports robustness and falsification tests, and Section 7 concludes. Additional results are in the Appendix and the Online Appendix.

2 Background and Conceptual Framework

2.1 Amazon’s Distribution Network

Amazon’s logistics network has three layers. Fulfillment centers store inventory and pick and pack orders, sortation centers group packages by destination, and delivery stations handle the last mile to customers (Cunningham, 2024). Air hubs and delivery stations for heavy and bulky items complete the network. Amazon operated a small number of fulfillment centers in 2010 and more than one thousand facilities by the early 2020s (Figure 1). The period I study, 2010 through 2024, covers the fastest phase of this expansion.

Amazon’s location decisions changed during this period. Through the 2000s, the company placed fulfillment centers in a small number of low-population states, in part to avoid collecting sales tax in the large states where most of its customers lived (Houde et al., 2016). Legal challenges and new state legislation made this strategy unsustainable. From about 2012, Amazon located facilities to minimize shipping costs and delivery times, which required building near customers in large, high-tax states (Chung, 2023; MWPVL International, 2025). The result was a staggered entry into local labor markets over more than a decade, which provides the variation in timing that this paper uses.

The treatment in this paper is Amazon’s entry into a commuting zone, defined as the opening of its first facility of any type. Entry usually brings more than one facility over time: of the 188 commuting zones Amazon entered, 59 received a fulfillment center in the year of entry and 112 had one by 2024, while in most of the rest the first facility was a delivery station or a sortation center. The estimates therefore capture the effect of Amazon’s arrival as a whole rather than of a single type of facility.¹

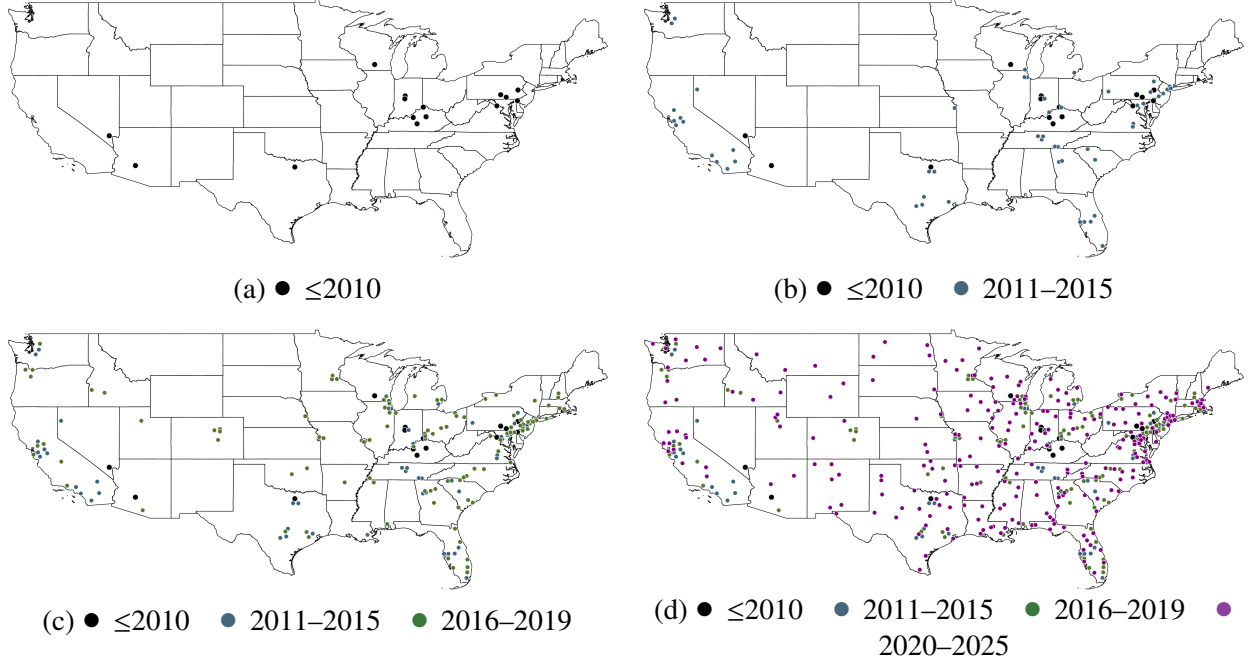
2.2 Conceptual Framework

I organize the analysis around a simple model of schooling choice in a local labor market, following Becker (1964) and the college-enrollment framework of Charles et al. (2018). A young person i with academic ability $\theta_i \in [0, 1]$ chooses among three options: enroll in a four-year college ($c = B$), enroll in a two-year college ($c = A$), or work without further schooling. The net payoff from college type c is

$$R^c(\theta_i) = \mathbb{E}[\Pi_c] - F_c - \kappa_c(1 - \theta_i) - Y_{NC}, \quad (1)$$

¹Amazon does not choose locations at random. Chung (2023) finds that fulfillment centers are more likely to open in counties with growing population density and relatively low median household income, consistent with Amazon seeking large labor pools close to major consumer markets. Section 3.2 discusses how the design addresses this selection.

Figure 1: Staggered Rollout of Amazon’s Logistics Network by Year, 2005–2025



Notes: Each panel plots the cumulative distribution of the county locations of Amazon facilities of all types (fulfillment centers, sortation centers, delivery stations, and air hubs) by opening-year cohort: black dots, facilities opened by 2010; blue dots, 2011–2015; green dots, 2016–2019; purple dots, 2020–2025. Data compiled from MWPVL International facility-tracking records. Facility locations are shown at the county level; the empirical analysis is conducted at the commuting-zone level.

where $\mathbb{E}[\Pi_c]$ is the expected lifetime earnings gain from degree c , F_c is the direct cost of attending, $\kappa_c(1 - \theta_i)$ is an effort cost that declines with ability, and Y_{NC} is the earnings forgone while in school, measured by the wages available to workers without a degree. A labor demand shock S , here the arrival of Amazon, changes the payoff by

$$\frac{dR^c(\theta_i)}{dS} = \underbrace{\frac{d\mathbb{E}[\Pi_c]}{dS}}_{\text{upgrading}} - \underbrace{\frac{dF_c}{dS}}_{\text{income}} - \underbrace{\frac{dY_{NC}}{dS}}_{\text{substitution}}. \quad (2)$$

Substitution lowers the payoff to college. Amazon’s warehouses hire workers without a degree. To the extent that they raise the wages or the availability of such work, they raise Y_{NC} and reduce it, the channel documented for coal, fracking, housing, and export-manufacturing booms (Black et al., 2005; Cascio and Narayan, 2022; Charles et al., 2018; Atkin, 2016). The substitution channel is strongest for students on the margin between work and school, such as those who combine work with part-time study at two-year colleges.

Upgrading works in the opposite direction. Amazon’s facilities also employ technical and

managerial staff, and they can draw suppliers and business services that employ college graduates (Greenstone et al., 2010; Rossi-Hansberg et al., 2021). It is useful to write the expected gain from a degree as the product of two terms,

$$\mathbb{E}[\Pi_B] = p_B \cdot \Delta w_B, \quad (3)$$

where p_B is the probability that a graduate finds a job in the local labor market that uses the degree and Δw_B is the wage gain in such a job. A demand shock can raise the expected gain through either term. If it raises the college wage premium Δw_B , the local return to a degree rises in price. If it raises the number of skilled jobs, and therefore p_B , the return rises even when the wage premium does not. The second case is especially likely when college graduates are mobile: new skilled jobs are then filled partly by graduates who stay in or move to the area, which keeps the local premium from rising.

Income works through the cost of college. If Amazon's entry raises household income and families face borrowing constraints, the effective cost F_c falls (Lochner and Monge-Naranjo, 2011; Lovenheim, 2011; Carneiro and Heckman, 2002; Johnson, 2013). This channel should matter most where constraints bind most tightly, that is, in poorer places.

Beyond schooling decisions, the stock of degree holders in a commuting zone can also change because graduates move. The same increase in skilled jobs that raises p_B for local students makes the area more attractive to graduates elsewhere and less attractive to leave (Notowidigdo, 2020; Monras, 2018). The two margins differ in timing: a change in schooling decisions reaches the stock of adults with a degree only after the years needed to complete it, whereas a change in location decisions can affect it almost immediately.

The framework yields four predictions that guide the empirical analysis. First, if upgrading dominates substitution, four-year enrollment and bachelor's attainment rise, while enrollment among students who combine work with part-time or two-year study may fall. Second, if upgrading operates through the availability of skilled jobs, the shares of employment in skilled industries and occupations rise, while the college wage premium need not. Third, if the income channel matters, effects are larger where households were poorer before Amazon arrived. Fourth, if migration matters, the stock of degree holders responds faster than new enrollment alone can explain. Whether the net effect on education is positive or negative is an empirical question.

3 Data and Empirical Strategy

3.1 Data

I combine data on the timing and location of Amazon’s facilities with data on the education, enrollment, and labor markets of the places they enter. Information on Amazon’s facilities comes from MWPVL International ([MWPVL International, 2025](#)), a supply-chain consulting firm that tracks Amazon’s logistics network. For each facility, the data record the address, the opening date, the facility type (fulfillment center, sortation center, delivery station, or air hub), and the floor space. I use the five facility types in Amazon’s logistics network and do not include Amazon Fresh or Whole Foods grocery stores, which are retail outlets rather than logistics facilities. I assign each facility to its 2020 commuting zone and define a commuting zone’s entry year as the year its first facility opened.

The main outcomes come from published American Community Survey 1-year estimates for 2005–2024, which the Census Bureau releases for every county with at least 65,000 residents; 2020 is missing because standard 1-year estimates were not released for that year. For adults aged 25 and older, I compute the shares with at least a high school diploma, some college, a bachelor’s degree, and a master’s degree, overall and by gender, and the bachelor’s share by race and ethnicity. I sum the population counts in each category across the counties of each 2020 commuting zone before computing shares; the few county-years in which the Census Bureau suppresses a table are filled by interpolation from the same county’s adjacent years. Because the estimates cover only larger counties, the panel describes the more populous part of each commuting zone.

College enrollment comes from the Integrated Postsecondary Education Data System (IPEDS) Fall Enrollment survey for 2005–2023. I keep one record per institution and year so that each student is counted once. The main enrollment outcome is the share of students enrolled part time. Following [Charles et al. \(2018\)](#), I also measure first-time enrollment of degree-seeking undergraduates, separately for four-year and two-year institutions, as a percentage of the commuting zone’s population aged 18–24. Because a college’s enrollment includes students from other places, I also construct enrollment at local-draw colleges, which exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. IPEDS also reports, from 2012, enrollment in distance education.

To study why attainment changes, I draw on three further sources. Household income, rents, poverty, unemployment, labor force participation, employment by industry and occupation, hourly wages, and migration status (whether a person lived in the same state one year earlier) come from IPUMS American Community Survey 1-year microdata. The ACS identifies Public Use Microdata Areas (PUMAs), which I allocate to counties with population-weighted allocation factors, using the crosswalk that matches each year’s PUMA definitions (2000 PUMAs for 2005–2011, 2010 PUMAs

for 2012–2021, and 2020 PUMAs for 2022–2024), and then to 2020 commuting zones. Because the allocation factor is constant within a PUMA, I first sum person weights within PUMA-year cells and then allocate the cells, which reproduces the person-level allocation exactly. Employment by 2-digit NAICS sector comes from County Business Patterns (CBP), 2005–2023; for years before 2017, cells withheld for disclosure are set to the midpoint of the reported employment size class. Establishment counts by sector come from the Census Bureau’s Business Dynamics Statistics (BDS), 2005–2023. County-to-county migration flows and the income of movers and non-movers come from the IRS Statistics of Income migration data, 2011–2023.

I aggregate all county-level data to 2020 commuting zones, so that each unit represents a single local labor market. The end result is a panel of 297 commuting zones observed in every year from 2005 to 2024, of which 188 receive an Amazon facility between 2010 and 2024. As an independent check on the attainment results, I also construct a second panel of attainment shares from the IPUMS microdata. It covers all 577 commuting zones observed in every year, including those without a county of 65,000 residents, of which 204 receive an Amazon facility, and is analyzed in Online Appendix Section [OA4](#).

3.2 Empirical Specification

Comparing the education of places with and without Amazon facilities is likely to give biased estimates of Amazon’s effect. Amazon entered large, prosperous, and educated places first, so a cross-sectional comparison would attribute to Amazon differences that existed before it arrived. Rather than relying on this variation, I exploit differences in the timing of entry: I compare the change in outcomes in commuting zones Amazon has already entered with the change over the same years in commuting zones it will enter later.

Let G_c denote the year in which commuting zone c is first treated. To allow for anticipation, for example hiring and construction before a facility opens ([Chung, 2023](#)), I set G_c to the year before the first facility opens. Treatment is absorbing: once Amazon enters, it stays, and the estimates capture the cumulative effect of all facilities that follow the first. The event-study specification is

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{e \neq -1} \theta_e \mathbf{1}[t - G_c = e] + \varepsilon_{ct}, \quad (4)$$

where Y_{ct} is an outcome, such as the share of adults with a bachelor’s degree, in commuting zone c and year t , α_c and λ_t are commuting-zone and year effects, and θ_e is the effect e years after treatment, relative to the year before treatment.

Amazon’s entry is staggered over fifteen years, and its effects grow over time. In this setting, estimating equation (4) by ordinary least squares compares newly treated commuting zones with

commuting zones treated earlier and can assign negative weights to some treatment effects (Goodman-Bacon, 2021; Sun and Abraham, 2021; De Chaisemartin and d’Haultfœuille, 2020; Borusyak et al., 2021; Roth et al., 2023). I therefore estimate the coefficients with the estimator of Callaway and Sant’Anna (2021). Its building block is the effect for the cohort of commuting zones first treated in year g , measured in year t ,

$$ATT(g, t) = \mathbb{E} [Y_{ct} - Y_{c, g-1} \mid G_c = g] - \mathbb{E} [Y_{ct} - Y_{c, g-1} \mid G_c > t], \quad (5)$$

the change in the outcome since the year before treatment in cohort g , minus the same change in commuting zones not yet treated in year t . I estimate each $ATT(g, t)$ with the doubly-robust estimator, which equals equation (5) when no covariates are included. The event-study coefficients average these effects across cohorts at each event time,

$$\theta_e = \sum_g \Pr(G_c = g \mid G_c + e \leq T) ATT(g, g + e), \quad (6)$$

and, because the effects build up over the first seven years after entry, the headline estimate is their average over $e = 0, \dots, 6$,

$$\bar{\theta} = \frac{1}{7} \sum_{e=0}^6 \theta_e. \quad (7)$$

The main estimation sample consists of the commuting zones that eventually receive a facility, so the comparison group is always made of places Amazon enters later. Restricting the sample in this way keeps the comparison among places Amazon judged suitable for a facility. The enrollment estimates are the exception: because many commuting zones lack colleges of a given type in some years, their comparison group also includes commuting zones Amazon never enters. Time-varying local characteristics are themselves affected by Amazon’s entry and are not used as controls. The main estimates use no covariates. In the robustness specifications with controls, the estimator conditions on two baseline characteristics measured before any commuting zone is treated: log population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009, the dimensions on which early and later entrants differ most.

My empirical strategy therefore compares the change in educational attainment in a commuting zone after Amazon enters with the change over the same years in not-yet-treated commuting zones, those Amazon enters later, which provide the counterfactual for how attainment would have evolved without Amazon.

3.3 Threats to Identification

The identifying assumption is that, absent Amazon’s entry, the outcomes of commuting zones Amazon enters earlier would have evolved like those of commuting zones it enters later. I address three concerns.

The first is selection on levels. Amazon enters large, prosperous, and educated places first. Because the estimator compares changes, and because the comparison group consists of places Amazon eventually enters, differences in levels do not bias the estimates.

The second is selection on trends. Early entrants may have been on a different path before Amazon arrived, for instance if Amazon chose places where education was already rising. I test this in three ways: balance tests on pre-treatment changes between early and later entrants ([Marcus and Sant’Anna, 2021](#)), the pre-treatment coefficients of the event studies, and placebo tests that move entry earlier (Sections [4.1](#) and [6](#)).

The third is a concurrent local shock, such as a rezoning or a public investment made to attract Amazon, that coincides with Amazon’s entry. Because such actions usually accompany entry, the estimates capture the effect of Amazon’s arrival together with the local changes that come with it, which is the relevant quantity for a place deciding whether to attract a facility. The content of the estimated changes is consistent with Amazon’s own demand for skills: the growth is concentrated in professional and technical services and in skilled occupations (Section [5.2](#)).

Finally, inference uses the multiplier bootstrap with 1,000 draws, clustered at the commuting-zone level to allow for serial correlation within a commuting zone, and the event-study figures show uniform confidence bands, which cover all event-time coefficients jointly.

4 Results

I first describe the commuting zones Amazon enters and test whether early and later entrants were on similar paths before entry. I then report the effects on educational attainment, on the composition of the adult population, and on college enrollment. Section [5](#) examines mechanisms.

4.1 Descriptive Statistics and Balance

Table [1](#) describes the commuting zones in the sample in 2005–2009, before Amazon’s expansion. Amazon enters large, urban, and relatively prosperous places. The average commuting zone that receives a facility has 1.22 million residents, about 8 times as many as a commuting zone Amazon never enters (Appendix Table [A1](#)); its median household income is \$68,939, against \$60,371; and 27.2 percent of its adults hold a bachelor’s degree, against 23.4 percent. Its labor market is also more skilled: college graduates hold 27.0 percent of jobs and STEM occupations account for 4.6

percent of employment, against 21.8 and 3.2 percent. Transportation and warehousing, Amazon's own sector, employs only a slightly larger share of workers in treated commuting zones (3.7 against 3.4 percent), so Amazon did not simply locate where logistics was already concentrated. Entry accelerated late in the period: 66 percent of treated commuting zones received their first facility in 2018 or later, and by 2024 a treated commuting zone had 6.3 facilities on average.

These differences in levels are why the design compares commuting zones Amazon has already entered with those it enters later, rather than with places it never enters. What matters for identification is whether early and later entrants were changing at the same rate before entry. Following [Marcus and Sant'Anna \(2021\)](#) and [Cunningham \(2024\)](#), I group treated commuting zones by the year of their first facility and compare each group with the commuting zones entered after it, using only years before treatment. Early entrants are larger and more educated in levels (Appendix Table [A3](#)), but in 40 of 44 comparisons their annual changes before entry do not differ significantly from those of later entrants (Appendix Table [A2](#)). All four exceptions are in the 2014–2016 group, whose shares with a high school diploma and with some college were rising more slowly, and whose college-graduate share of employment and unemployment rate were rising faster, than those of later entrants.

4.2 Educational Attainment

Figure [2](#) reports the event-study estimates for the four attainment measures and Table [2](#) the aggregated effects. Amazon's entry raises the share of adults with a bachelor's degree by 0.48 percentage points (standard error 0.18), about 2 percent of the pre-entry mean of 28.0 percent, and the share with a master's degree by 0.39 percentage points (standard error 0.10), about 4 percent of its mean. The estimates for a high school diploma and for some college are small and insignificant. The effects build up over time: the bachelor's share is unchanged in the year of entry, rises by 0.21 points two years later, and reaches 1.3 points after six years; the master's share reaches 0.8 point after six years. The event-study coefficients become individually significant from the fourth year after entry.

The pre-treatment coefficients are close to zero for all four outcomes. The gains are broad-based: they are similar for men and women and precisely estimated for White adults, while estimates for smaller groups are imprecise (Appendix Table [A4](#)).

4.3 Migration and the Composition of the Adult Population

A higher share of adults with a degree can reflect more residents earning degrees or more degree holders moving in. Table [3](#) examines both, using the same specification as the attainment results.

Panel A shows that the share rises because the number of graduates grows. In the published ACS estimates, the number of adults with a bachelor's degree rises by 2.3 percent, about twice as fast as the number without one, which rises by 1.2 percent. Place of birth, which does not change

Table 1: Education, Enrollment, and Labor Markets Before Amazon’s Expansion

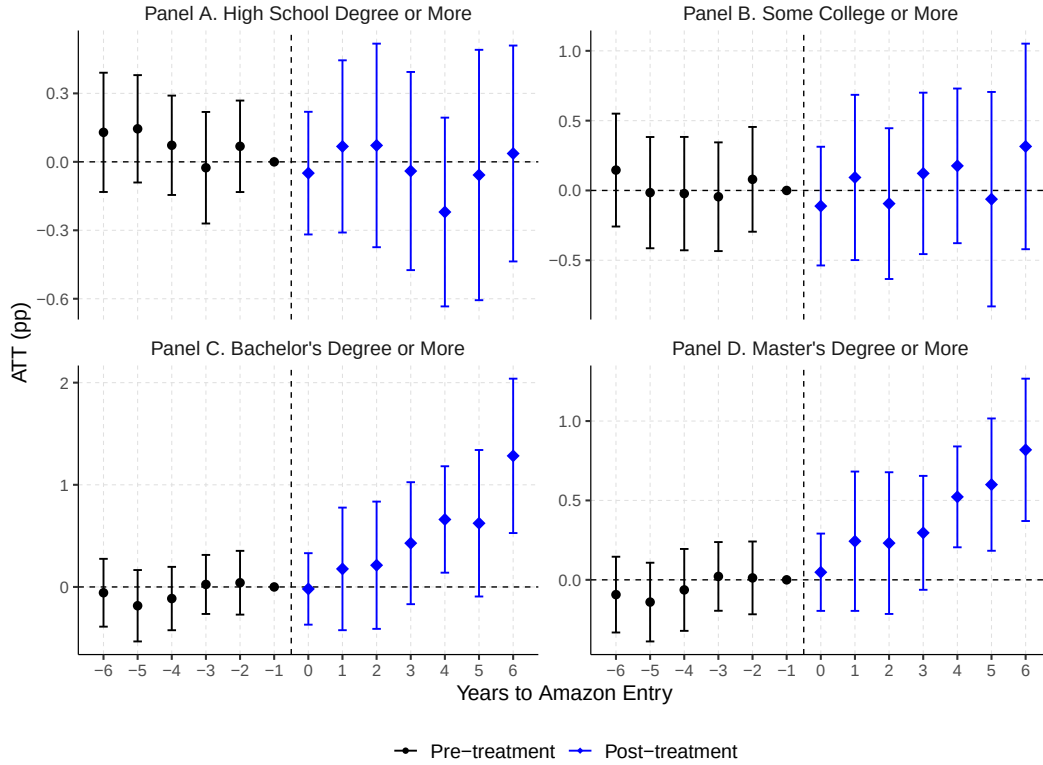
	All commuting zones (1)	Receive Amazon facility (2)	Never receive facility (3)	Difference (2) – (3) (4)
<i>Panel A. Educational attainment, adults aged 25+ (%)</i>				
High school diploma or more	85.9 (4.8)	86.2 (4.4)	85.5 (5.3)	0.7 [0.6]
Some college or more	55.3 (7.8)	56.5 (6.9)	53.0 (8.8)	3.5*** [1.0]
Bachelor’s degree or more	25.8 (7.5)	27.2 (6.6)	23.4 (8.3)	3.9*** [0.9]
Master’s degree or more	9.4 (3.6)	9.8 (3.2)	8.5 (4.0)	1.3*** [0.4]
<i>Panel B. College enrollment (IPEDS)</i>				
First-time students, four-year colleges (% of ages 18–24)	3.84 (3.36)	4.11 (3.12)	3.38 (3.72)	0.73* [0.42]
First-time students, two-year colleges (% of ages 18–24)	4.25 (2.23)	4.20 (1.84)	4.33 (2.79)	–0.13 [0.30]
Part-time students (% of enrollment)	36.9 (12.1)	36.0 (10.9)	38.4 (13.8)	–2.4 [1.5]
<i>Panel C. Income and labor market</i>				
Median household income (2023 \$)	65,795 (12,084)	68,939 (12,465)	60,371 (9,179)	8,569*** [1,265]
Poverty rate (%)	13.1 (3.9)	12.5 (3.6)	14.1 (4.1)	–1.6*** [0.5]
Unemployment rate (%)	7.1 (1.5)	7.0 (1.4)	7.2 (1.8)	–0.2 [0.2]
Labor force participation rate (%)	64.1 (4.5)	65.2 (4.0)	62.4 (4.8)	2.8*** [0.5]
College graduates (% of employment)	25.1 (6.4)	27.0 (6.2)	21.8 (5.3)	5.3*** [0.7]
STEM occupations (% of employment)	4.1 (1.8)	4.6 (1.8)	3.2 (1.2)	1.4*** [0.2]
Abstract-task occupations (% of employment)	32.2 (4.1)	33.4 (4.0)	30.2 (3.5)	3.2*** [0.4]
Professional & technical services (% of employment)	4.8 (2.4)	5.5 (2.5)	3.7 (1.5)	1.7*** [0.2]
Transportation & warehousing (% of employment)	3.6 (1.5)	3.7 (1.5)	3.4 (1.5)	0.3* [0.2]
Commuting zones	297	188	109	

Notes: Each commuting zone’s characteristics are averaged over 2005–2009, before any commuting zone in the sample is treated, and the table reports the mean across commuting zones, with standard deviations in parentheses; commuting zones are not weighted by population. Column (2) contains commuting zones where Amazon opened at least one facility between 2010 and 2024, column (3) the remaining commuting zones in the balanced 2005–2024 panel. Column (4) reports the difference in means, with its standard error in brackets; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Attainment is from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to 2020 commuting zones; income, poverty, labor-market, and occupation measures are from IPUMS ACS microdata for the same commuting zones; household income is converted to 2023 dollars with the CPI-U. Enrollment is from IPEDS (first-time students, 2006–2009): first-time degree-seeking undergraduates at local-draw colleges as a percentage of the population aged 18–24, and part-time students as a percentage of all students (commuting zones with colleges). STEM occupations combine computer and mathematical, engineering, and science occupations; abstract-task occupations are the top employment-weighted third of the [Autor and Dorn \(2013\)](#) abstract task index. Industry shares are from County Business Patterns.

over a lifetime, shows whether the increase is concentrated among people who came from elsewhere. The number of graduates born in their state of residence rises by 2.2 percent and the number born in another state by 2.1 percent, so graduates from other states do not grow faster than local ones. The number of foreign-born graduates rises faster, by 6.1 percent.

Panel B turns to recent moves, which show that Amazon’s entry does attract educated movers. The number of adults who moved from another state or from abroad in the past year does not change, but its composition shifts toward graduates. Recent in-migrants with a bachelor’s degree rise by 0.066 per 100 adults, about 7 percent of their pre-entry level, and those with a master’s degree

Figure 2: Event-Study Estimates: Effects of Amazon’s Entry on Educational Attainment, 2005–2024



Notes: Each panel plots event-time coefficients; black circles are pre-treatment and blue diamonds post-treatment coefficients. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone (to allow for anticipation). Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant’Anna \(2021\)](#), comparing treated commuting zones to not-yet-treated controls.

by 0.040, about 11 percent, while recent in-migrants without a degree do not change. None of these series differs significantly before entry. The inflow is nevertheless modest relative to the change in the stock. If every additional graduate who moved in stayed, the cumulative inflow would average 0.21 graduates per 100 adults over the post-entry years, against the 0.48-point increase in the bachelor’s share in the main estimates. In-migration from other states and abroad can therefore account for less than half of the increase. Consistent with this, the bachelor’s share rises by 0.69 percentage points among adults who did not move in the past year.

Panel C uses IRS records, which cover all movers but not their education. The number of people moving in does not change, and the number moving out rises slightly (1.05 per 1,000 residents, significant at the 10 percent level). Incomes rise by similar amounts for households that move in

Table 2: Effects of Amazon’s Entry on Educational Attainment

	High School+ (1)	Some College+ (2)	Bachelor’s+ (3)	Master’s+ (4)
Amazon Entry	−0.027 (0.129)	0.063 (0.153)	0.481*** (0.180)	0.394*** (0.104)
Mean Dep. Var.	87.54	58.46	27.99	10.29
SD	4.59	7.14	6.60	3.28
Commuting zones	188	188	188	188
N	3,572	3,572	3,572	3,572

Notes: Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant’Anna \(2021\)](#), averaged over the post-entry event years, comparing treated commuting zones to not-yet-treated controls. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens (anticipation). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. and SD report the pre-treatment mean and standard deviation in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(2.6 percent) and households that stay (2.9 percent), so the income records add no evidence either way on the education of movers.

Amazon’s entry therefore draws more graduates to these commuting zones, from other states and from abroad, but these newcomers explain less than half of the rise in the bachelor’s share. Most of the increase comes from people who already live there, either residents who complete degrees or graduates who stay rather than leave. The data cannot separate these two groups, because they do not follow the same people over time and do not record moves within a state.

4.4 College Enrollment

Besides new students who enroll after Amazon arrives, two responses can raise the share of adults with a degree. The first is migration. As [Section 4.3](#) suggests, part of the increase may come from graduates who stay in or move to commuting zones Amazon has entered. The second is faster completion by students who are already enrolled, since a student who switches from part-time to full-time study can finish a degree sooner. The enrollment data provide suggestive evidence of this second response.

Amazon’s entry changes how students attend. Unlike the attainment estimates, the enrollment estimates in [Table 4](#) and [Appendix Table A5](#) average the post-treatment effects across treatment cohorts. In commuting zones that receive a facility, the share of students enrolled part time falls by 1.15 percentage points, and the decline widens to 3.7 points after six years ([Table 4](#) and [Figure 3](#)). The decline combines two responses: full-time enrollment grows at public four-year colleges, and

Table 3: Amazon’s Entry, Migration, and the Composition of the Adult Population

	Pre-entry mean	Effect of Amazon’s entry	Commuting zones
<i>Panel A. Adults by degree and place of birth, published ACS estimates (log, effect in percent)</i>			
Adults without a bachelor’s degree	439.0	1.17** (0.57)	188
Adults with a bachelor’s degree	191.5	2.32*** (0.79)	188
born in state of residence	78.4	2.16** (1.05)	188
born in another state	79.5	2.14* (1.11)	188
foreign-born	30.7	6.08*** (2.32)	188
<i>Panel B. Adults who moved from another state in the past year, per 100 adults, ACS microdata</i>			
All	2.69	0.089 (0.056)	204
Without a bachelor’s degree	1.75	0.024 (0.042)	204
With a bachelor’s degree	0.94	0.066** (0.030)	204
With a master’s degree	0.37	0.040*** (0.015)	204
Bachelor’s share, adults who did not move (pp)	25.07	0.69*** (0.15)	204
<i>Panel C. Migration flows and household income, IRS records (2011–2023)</i>			
In-migrants per 1,000 residents	48.15	0.18 (0.50)	195
Out-migrants per 1,000 residents	48.00	1.05* (0.55)	197
Income of in-migrant households	32.9	2.63** (1.18)	195
Income of households that did not move	43.5	2.86*** (0.59)	203
In-migrant income, % of non-movers’	76.25	−0.05 (0.90)	195

Notes: Each row reports the average post-entry effect of Amazon’s entry, estimated as in Table 2: the doubly-robust estimator of Callaway and Sant’Anna (2021) with not-yet-treated comparison commuting zones, no covariates, and treatment in the year before the first facility opens. Panel A uses the published ACS 1-year county estimates of the main analysis: counts of adults aged 25 and older by degree (table B15002) and of adults with a bachelor’s degree by place of birth (table B06009). Pre-entry means of these counts are in thousands of persons, and effects are in percent (100 times log points). Panel B uses IPUMS ACS microdata, which cover all commuting zones. Its first four rows count adults aged 25 and older who lived in another state or abroad one year earlier, per 100 adults aged 25 and older; adults who did not move lived in the same state one year earlier. Panel C uses IRS Statistics of Income county-to-county migration data for 2011–2023; migrants are counted as exemptions, income is adjusted gross income per return in thousands of 1999 dollars (pre-entry means in levels, effects in percent), and commuting zones treated before 2012 have no pre-entry year and drop out. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

part-time enrollment falls at public two-year colleges, consistent with the substitution force of the framework for students who combine work and study. Before the pandemic, the share of students studying entirely online at local public colleges also fell (Online Appendix Table [OA4](#)).

The shift toward full-time study also appears among new students. I measure new enrollment following [Charles et al. \(2018\)](#): first-time, degree-seeking undergraduates at four-year and at two-year colleges in the commuting zone, as a percentage of the population aged 18–24. Because IPEDS records enrollment where a college is located, I focus on local-draw colleges, which exclude institutions that drew 30 percent or more of their first-time students from other states or abroad in 2006 and 2008; this exclusion removes 34 percent of four-year and 3 percent of two-year first-time enrollment. First-time enrollment at local four-year colleges rises by 0.215 percentage points (standard error 0.117), about 4 percent of the pre-entry mean of 5.08, and the effect at public four-year colleges grows to 0.51 points six years after entry (Appendix Table [A5](#) and Figure 4). The increase is concentrated among full-time students (0.185, standard error 0.095). Enrollment at two-year colleges does not change (−0.044). These estimates are less precise than the attainment results and they are significant at the 10 percent level.

Amazon’s entry therefore appears to shift enrollment toward full-time, in-person study at four-year colleges rather than pulling students out of school. Because full-time students tend to finish degrees faster, this shift may also add to attainment.

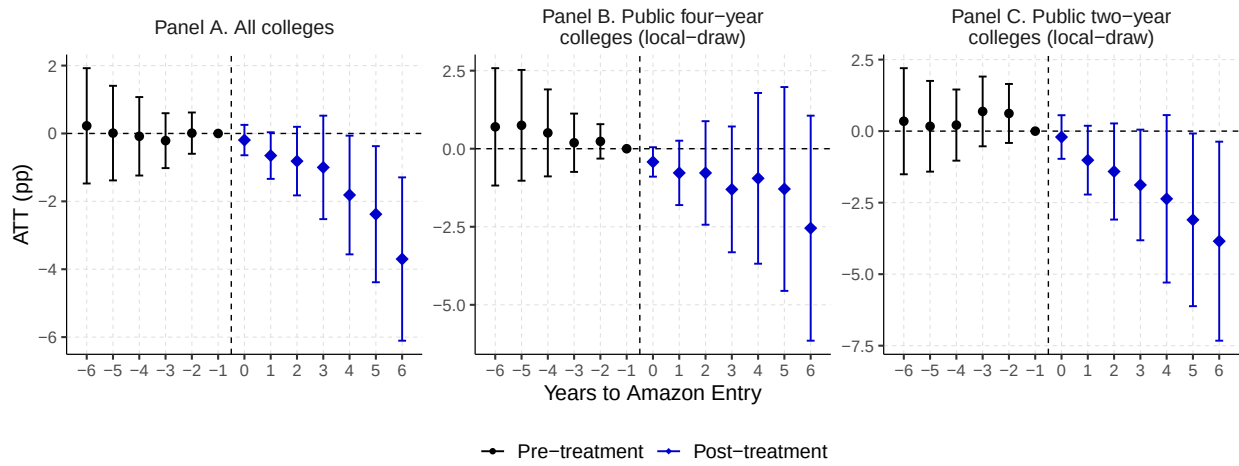
Table 4: Part-Time Share of Enrollment, 2010–2023

	All colleges (1)	Public four-year (local-draw) (2)	Public two-year (local-draw) (3)
Amazon Entry	−1.150** (0.463)	−0.974* (0.582)	−1.594*** (0.574)
Commuting zones	192	159	170
Mean Dep. Var.	36.40	28.76	56.22

Notes: Dependent variable is part-time enrollment as a percentage of total enrollment (all students, undergraduate and graduate) at colleges of the specified type located in the commuting zone, 2010–2023, from IPEDS Fall Enrollment. Four-year colleges are bachelor’s-granting institutions and two-year colleges do not award bachelor’s degrees. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. A commuting zone is treated from the year before its first Amazon facility opens; cohorts treated in 2010 have no pre-period and do not contribute. The sample is restricted to commuting zones that receive an Amazon facility, so commuting zones treated later serve as comparisons for those treated earlier. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant’Anna \(2021\)](#), aggregated across treatment cohorts; the estimator differences out time-invariant commuting-zone characteristics and common year shocks. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones.

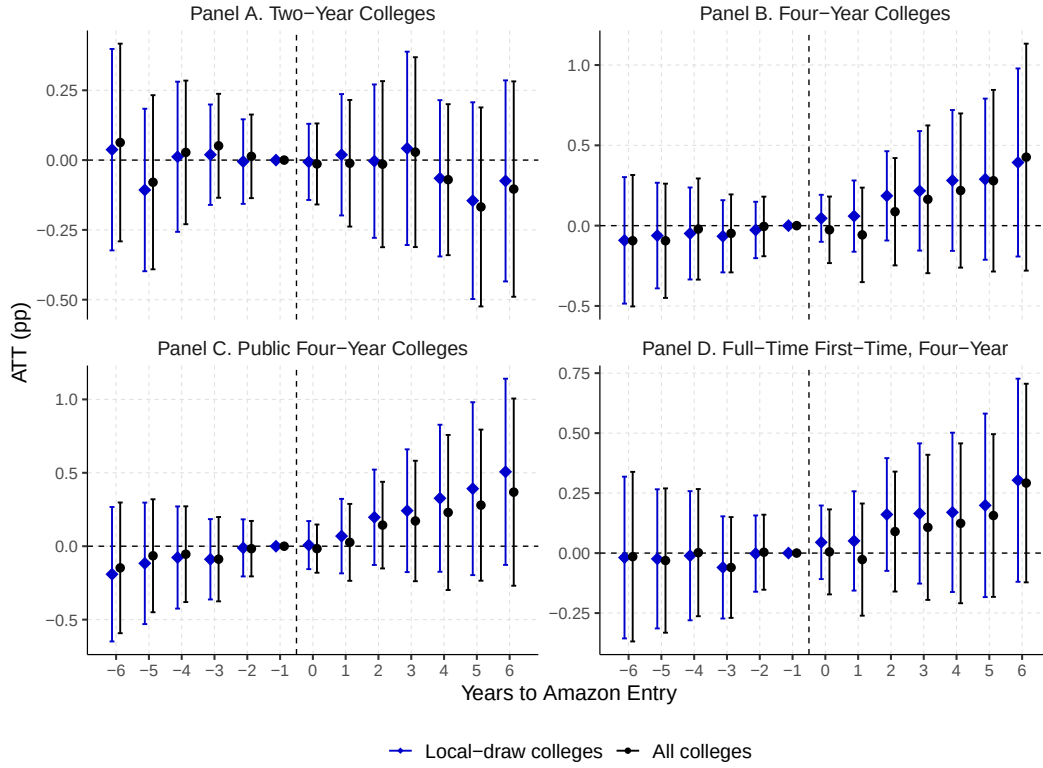
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3: Event-Study Estimates: Part-Time Share of Enrollment, 2010–2023



Notes: Each panel plots event-time coefficients for part-time enrollment as a percentage of total enrollment at colleges of the specified type in the commuting zone. The sample is restricted to commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons (Table 4). Black circles are pre-treatment and blue diamonds post-treatment coefficients. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon's first facility opens in the commuting zone. Enrollment is from IPEDS Fall Enrollment, 2010–2023. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant'Anna \(2021\)](#).

Figure 4: Event-Study Estimates: First-Time College Enrollment, 2010–2023



Notes: Each panel plots event-time coefficients for first-time enrollment as a percentage of the commuting zone’s population aged 18–24, estimated separately for local-draw colleges (blue diamonds), which exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008, and for all colleges (black circles). Panels A–C use first-time degree- or certificate-seeking undergraduate enrollment; Panel D uses full-time first-time enrollment at four-year colleges. Two-year colleges do not award bachelor’s degrees; four-year colleges are bachelor’s-granting institutions (Charles et al., 2018). Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Enrollment is from IPEDS Fall Enrollment (2010–2023) aggregated to 2020 commuting zones; population is from IPUMS ACS 1-year microdata. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of Callaway and Sant’Anna (2021), comparing treated commuting zones with not-yet-treated.

5 Mechanisms

The conceptual framework offers two reasons why Amazon’s entry could raise education despite the substitution force of its warehouse jobs: higher household income, which relaxes financial constraints, and higher demand for skilled work, which raises the expected return to a degree.

5.1 Household Income

Median household income rises gradually after Amazon’s entry, by 1.4 percent on average, or about \$920 per household in 2023 dollars, and by 3.9 percent six years after entry (Appendix Table A6, Panel A). The mean wage of workers does not change, and the unemployment rate falls modestly. The poverty rate does not change. Rents rise faster than income, by 3.7 percent on average. If the education effects worked through relaxed credit constraints, they should be largest where households were poorest before Amazon arrived. They are not. Splitting treated commuting zones at the median of the 2005–2009 poverty rate, household income rises by 2.8 percent in lower-poverty commuting zones and not at all in higher-poverty ones, and the bachelor’s-degree effect is also larger in the lower-poverty half (0.56 against 0.20 percentage points; Appendix Table A6, Panel B). The income channel is therefore unlikely to explain the attainment results.

5.2 The Demand for Skilled Work

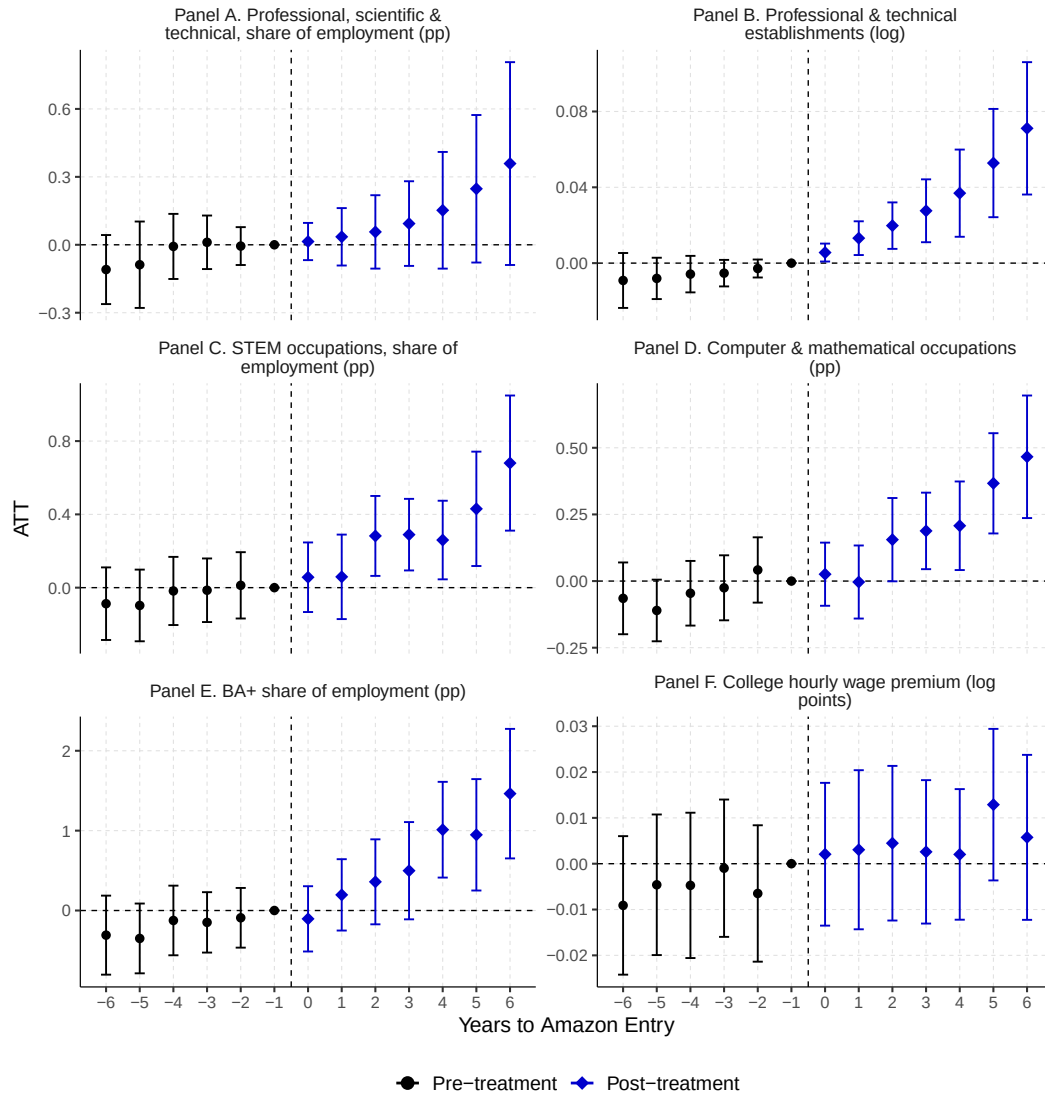
Amazon’s facilities are, first of all, warehouses, and most of their own jobs do not require a degree. Figure 5 and Table 5 show that Amazon’s arrival nevertheless shifts local employment toward skilled work. Because Amazon locates in growing areas (Chung, 2023), the analysis uses industry and occupation shares, which net out common growth; none of the outcomes reported shows a pre-treatment trend.

The first change is in the mix of businesses. The share of employment in professional, scientific, and technical services, the sector of engineering, consulting, accounting, and computer services, rises by 0.14 percentage points on average and by 0.36 points six years after entry, and the number of such establishments grows by 3.3 percent, significantly in every year after entry (Appendix Table A9). The shares of retail trade, accommodation and food services, administrative support, and manufacturing fall. The second change is in occupations. The share of employment in STEM occupations rises by 0.29 percentage points from a base of 4.4 percent, driven by computer and mathematical occupations, whose share rises by about 10 percent of its pre-entry level; professional and managerial occupations gain 0.40 points, and the share of employment in abstract-task occupations, as defined by Autor and Dorn (2013), rises by 0.42 points. The share of clerical sales and office jobs changes by –0.48 points, and transportation and material moving occupations, which include warehouse work, rise

by only 0.17 points. The third change is in who holds the jobs: the share of employment held by college graduates rises by 0.62 percentage points.

The college wage premium, by contrast, does not change. Hourly wages of college graduates (0.000 log points) and of other workers (-0.005) are unchanged, and so is their difference (0.005, standard error 0.004). In terms of equation (3), Amazon's entry raises the expected return to a degree through p_B , the availability of skilled jobs, rather than through Δw_B , their pay. A flat premium alongside rising skilled employment suggests that the supply of graduates keeps pace with demand, through graduates who stay or arrive and through residents who shift toward full-time, four-year study. The occupation data cannot separate skilled jobs at Amazon itself from jobs at other employers, but the growth of professional and technical establishments shows that the change extends beyond Amazon's own operations.

Figure 5: Event-Study Estimates: Effects of Amazon’s Entry on Skilled Employment and Wages



Notes: Each panel plots event-time coefficients for the specified outcome. Panel A (professional, scientific, and technical services) is from County Business Patterns and Panel B from the Business Dynamics Statistics, 2005–2023, aggregated from counties to commuting zones; Panels C–F are from IPUMS ACS microdata, 2005–2024. STEM occupations combine computer and mathematical, architecture and engineering, and life, physical, and social science occupations. The college hourly wage premium is the difference in mean log hourly wages between workers aged 25–64 with and without a bachelor’s degree. Black circles are pre-treatment and blue diamonds post-treatment coefficients. The sample is the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant’Anna \(2021\)](#).

Table 5: Effects of Amazon’s Entry on the Demand for College-Educated Workers

	Amazon Entry (1)	Std. error (2)	Mean before entry (3)
<i>Panel A. Education of the workforce</i>			
BA+ share of employment (pp)	0.625***	(0.158)	27.24
<i>Panel B. Occupation share of employment (pp)</i>			
STEM occupations	0.294***	(0.060)	4.39
Computer & mathematical	0.201***	(0.043)	1.93
Professional & managerial	0.400**	(0.167)	33.29
Management, business & financial	0.237**	(0.104)	13.06
Abstract-task occupations (Autor–Dorn)	0.421***	(0.142)	33.17
Sales & office	−0.481***	(0.119)	23.95
Transportation & material moving	0.170**	(0.085)	6.78
<i>Panel C. Wages (log points)</i>			
Log hourly wage, BA+ workers	0.000	(0.004)	2.87
Log hourly wage, non-BA workers	−0.005	(0.003)	2.39
College hourly wage premium	0.005	(0.004)	0.47
College annual wage premium	−0.001	(0.006)	0.59

Notes: Outcomes are computed from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024. Panel A: bachelor’s-degree holders as a share of all employed persons aged 16 and older. Panel B: occupation groups of employed persons aged 16 and older, based on the harmonized 2010 occupation codes; STEM combines computer and mathematical, architecture and engineering, and life, physical, and social science occupations; professional and managerial adds management, business, and financial and other professional occupations; abstract-task occupations are those in the top employment-weighted third of the abstract task-intensity index of [Autor and Dorn \(2013\)](#) (occupations mapped to occ1990dd with the crosswalk of David Dorn; thirds set on national employment in 2005–2009). Panel C: hourly wages of workers aged 25–64 are annual wage and salary income divided by weeks worked (interval midpoints) times usual weekly hours, in constant 1999 dollars; the premium is the difference in mean log hourly wages between workers with and without a bachelor’s degree; the annual premium uses the log of mean annual earnings. A commuting zone is treated from the year before its first Amazon facility opens. Each row is a separate doubly-robust staggered difference-in-differences estimate ([Callaway and Sant’Anna, 2021](#)) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons; column (1) averages the event-time effects, and column (2) reports standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Column (3) is the pre-treatment mean in treated commuting zones (for log outcomes, the geometric mean count). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Robustness and Falsification

This section asks whether the attainment effects could arise without Amazon. Table 6 reports four checks. Panel A uses the main panel for a different estimator and a placebo test. Panel B uses the IPUMS microdata panel, which covers all commuting zones, for the inference and pandemic tests.

A different estimator gives similar results. The main estimates compare commuting zones Amazon has entered with those it enters later. The synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#) instead compares them with a weighted combination of commuting zones Amazon never enters, chosen to match their path before entry. On the main panel, it gives effects of 0.67 percentage points for the bachelor’s share and 0.45 for the master’s share, and 0.57 and 0.40

when the log adult population is projected out. These are similar to or somewhat larger than the main estimates of 0.48 and 0.39, and the pre-entry coefficients are close to zero (Appendix Figure A2). The synthetic estimator also finds a small decline in the share with a high school diploma (−0.25 percentage points; Online Appendix Section OA4), which the main estimates do not show.

The effects are also unlikely to arise by chance. In the microdata panel, the effect is 0.71 percentage points for the bachelor’s share and 0.54 for the master’s share, and 0.44 and 0.33 with baseline controls for population size and growth. I reassign Amazon’s entry years at random across the treated commuting zones 500 times, holding fixed the number of commuting zones entered in each year, and re-estimate the effect each time. The placebo estimates are centered on zero (Appendix Figure A1), and at most one of the 500 random assignments produces an effect as large as the actual one, for either outcome and either specification (randomization-inference $p \leq 0.002$).

A placebo test checks for pre-existing trends. I move each commuting zone’s entry five years earlier and estimate the effect of this fake entry on the main panel, using only years before Amazon actually arrived. The placebo effects are small and insignificant: 0.13 percentage points for the bachelor’s share ($p = 0.25$) and 0.06 for the master’s share ($p = 0.34$), and 0.19 and 0.08 with baseline controls ($p = 0.45$ and 0.49).

Finally, ending the sample in 2019, so that no pandemic year is used and commuting zones entered after 2019 serve as comparisons, leaves the results essentially unchanged: 0.68 percentage points for the bachelor’s share and 0.47 for the master’s share without controls, and 0.47 and 0.25 with controls.

Table 6: Robustness and Falsification Tests

	Bachelor's degree or more		Master's degree or more	
	No controls (1)	Baseline controls (2)	No controls (3)	Baseline controls (4)
<i>Panel A. Main panel: published ACS estimates, 188 treated commuting zones</i>				
Callaway–Sant’Anna (Table 2)	0.481*** (0.180)		0.394*** (0.104)	
Synthetic difference-in-differences	0.675*** (0.168)		0.446*** (0.113)	
log adult population projected out	0.566*** (0.176)		0.397*** (0.104)	
Placebo: entry moved five years earlier	0.133 (0.115)	0.186 (0.244)	0.064 (0.067)	0.084 (0.121)
<i>Panel B. IPUMS microdata panel, 204 treated commuting zones</i>				
Estimate, 2005–2024	0.706*** (0.141)	0.440** (0.185)	0.538*** (0.075)	0.326*** (0.096)
Randomization-inference p -value	<0.002	0.002	<0.002	<0.002
Pre-pandemic sample, 2005–2019	0.676*** (0.152)	0.471** (0.188)	0.474*** (0.081)	0.249** (0.115)

Notes: Each cell reports the average post-entry effect of Amazon’s entry, in percentage points of adults aged 25 and older. Panel A uses the published American Community Survey 1-year county estimates of the main analysis (2005–2024; 2020 is not available). Its first row repeats the main estimate from the doubly-robust estimator of [Callaway and Sant’Anna \(2021\)](#) with not-yet-treated comparisons. The synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#) uses the 109 commuting zones that never receive a facility as donors, without covariates or with the log adult population projected out; its standard errors come from 200 bootstrap replications. Panel B uses the IPUMS ACS microdata panel, which covers all commuting zones, and the estimator of [Callaway and Sant’Anna \(2021\)](#) with not-yet-treated comparisons. Baseline controls are log population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009. The randomization-inference p -value is the share of 500 random reassignments of treatment years across treated commuting zones, holding the number of commuting zones treated in each year fixed, that produce an estimate at least as large in absolute value as the actual one. The pre-pandemic sample ends in 2019, and commuting zones treated after 2019 serve as comparisons. The placebo, on the main panel, moves each commuting zone’s treatment date five years earlier and uses only years before the actual treatment (172 commuting zones whose fake treatment year is 2007 or later). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses, unless noted otherwise. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

Booms in low-skill work have long been found to pull young people out of school. This paper studies the largest recent expansion of such work in the United States, Amazon’s national rollout of logistics facilities, and finds the opposite. In the commuting zones Amazon enters, the share of adults with a bachelor’s degree rises by 0.48 percentage points and the share with a master’s degree by 0.39 points, and both effects grow over the following six years. The gains appear to arrive through two margins: graduates stay in or move to these commuting zones, and students shift from part-time to full-time

study. The results survive randomization inference, placebo tests that move Amazon’s entry earlier, the exclusion of the pandemic years, and a synthetic difference-in-differences estimator.

The evidence points to the demand for skilled work as the reason. After Amazon arrives, professional, scientific, and technical establishments expand, and employment shifts toward STEM, managerial, and abstract-task occupations. The college wage premium does not change. Amazon’s entry raises the number of skilled jobs available close to home rather than their pay, and the evidence suggests that both local residents and college graduates respond: the stock of degree holders rises as graduates stay in or move to the area, and enrollment shifts toward the full-time, four-year study that leads to these jobs. Household income rises only modestly, and mostly in commuting zones that were already better off, which makes relaxed financial constraints an unlikely explanation.

These findings bear on the debate over the incentives that states and localities offer to attract large employers. Evaluations of such incentives typically count jobs and wages. The results here suggest that the skill content of the jobs an employer brings, and of the activity that follows it, also shapes the education of the local population. Three limits apply. First, the estimates describe commuting zones, not individuals; individual data linking workers’ education histories to their employers and locations would separate new degrees from the location choices of graduates more precisely. Second, Amazon’s entry often coincides with local infrastructure investment and other development policies, which the design cannot fully separate from Amazon’s own effect. Third, whether other large logistics or technology employers generate similar effects, and whether the effects scale with the size of the incentives a locality offers, remain open questions.

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Appendix

Table A1: Population and Amazon's Presence

	All commuting zones (1)	Receive Amazon facility (2)	Never receive facility (3)	Difference (2) – (3) (4)
<i>Panel A. Population</i>				
Total population (thousands)	829 (1,616)	1,219 (1,926)	155 (110)	1,064*** [141]
Aged 18–24 (% of population)	10.9 (3.4)	10.6 (2.7)	11.4 (4.4)	–0.8 [0.5]
White, non-Hispanic (%)	75.2 (16.4)	72.2 (16.2)	80.5 (15.4)	–8.3*** [1.9]
Black (%)	10.2 (11.0)	11.7 (11.1)	7.8 (10.4)	3.9*** [1.3]
Hispanic (%)	9.0 (11.8)	10.6 (13.0)	6.1 (8.5)	4.6*** [1.3]
Foreign-born (%)	6.5 (6.2)	7.9 (6.9)	4.1 (3.7)	3.7*** [0.6]
Urban (% of population)	68.1 (16.8)	75.2 (14.1)	55.8 (13.7)	19.4*** [1.7]
<i>Panel B. Amazon's presence in treated commuting zones</i>				
Year first facility opened		2018.6 (3.8)		
Facilities in 2024 (median: 2)		6.3 (9.3)		
Facility floor space in 2024 (million sq. ft.)		2.04 (3.47)		
Commuting zones	297	188	109	

Notes: Each commuting zone's characteristics are averaged over 2005–2009, before any commuting zone in the sample is treated, and the table reports the mean across commuting zones, with standard deviations in parentheses; commuting zones are not weighted by population. Column (2) contains commuting zones where Amazon opened at least one facility between 2010 and 2024, column (3) the remaining commuting zones in the balanced 2005–2024 panel. Column (4) reports the difference in means, with its standard error in brackets; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Total population and the population aged 18–24 are from published ACS 1-year county estimates; race, ethnicity, and foreign birth are from IPUMS ACS microdata (2005–2009); urban share is from the 2010 Census. Panel B refers to commuting zones that receive a facility; the year of the first opening, the number of facilities, and their floor space in 2024 are from MWPVL International.

Table A2: Balance in Pre-Treatment Trends: Early versus Later Amazon Entry

	Year of first Amazon facility			
	2010–2013 (1)	2014–2016 (2)	2017–2019 (3)	2020–2021 (4)
High school or more (%)	–0.085 (0.067)	–0.060** (0.030)	–0.034 (0.027)	–0.012 (0.032)
Some college or more (%)	0.018 (0.111)	–0.095** (0.037)	–0.030 (0.036)	0.006 (0.044)
Bachelor’s or more (%)	0.025 (0.070)	0.043 (0.033)	0.044 (0.030)	0.051 (0.039)
Master’s or more (%)	0.038 (0.044)	0.032 (0.026)	0.034* (0.020)	0.026 (0.024)
First-time four-year students (% of 18–24)	0.068 (0.076)	0.013 (0.041)	0.011 (0.027)	0.001 (0.029)
College graduates (% of employment)	–0.007 (0.078)	0.129*** (0.039)	0.033 (0.033)	0.011 (0.038)
STEM occupations (% of employment)	0.076* (0.045)	0.011 (0.016)	–0.010 (0.014)	–0.008 (0.013)
Log median household income	0.0016 (0.0022)	–0.0026 (0.0017)	–0.0017 (0.0012)	–0.0021* (0.0012)
Poverty rate (%)	0.081 (0.088)	0.093* (0.050)	0.054 (0.036)	–0.004 (0.042)
Unemployment rate (%)	0.067 (0.068)	0.146*** (0.057)	0.015 (0.025)	0.007 (0.022)
Log population	0.0001 (0.0028)	–0.0012 (0.0024)	–0.0006 (0.0020)	0.0017 (0.0016)
Pre-treatment years	2005–2008	2005–2012	2005–2015	2005–2018
Commuting zones in the group	25	29	39	53
Comparison commuting zones	163	134	95	42

Notes: Each cell reports the difference in the average annual change of the variable during the pre-treatment years, between the commuting zones in the group and the comparison commuting zones, following [Marcus and Sant’Anna \(2021\)](#) and [Cunningham \(2024\)](#). A difference of zero means the two groups were on the same trend before Amazon’s entry. Each column compares commuting zones whose first Amazon facility opened in the indicated years with commuting zones that received their first facility later, which serve as their comparison group in the difference-in-differences design, using only years before the group is treated (treatment begins the year before the first opening). The sample is the balanced 2005–2024 panel (2020 not available) of commuting zones that receive an Amazon facility; the comparison group for the 2020–2021 group consists of commuting zones first entered in 2022–2024. Standard errors, allowing for unequal variances, in parentheses. Attainment and population are from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to commuting zones; income, poverty, unemployment, and workforce measures are from IPUMS ACS microdata; first-time four-year students (local-draw colleges, as a percentage of the population aged 18–24) are from IPEDS, available from 2006. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Balance in Pre-Treatment Levels: Early versus Later Amazon Entry

	Year of first Amazon facility			
	2010–2013 (1)	2014–2016 (2)	2017–2019 (3)	2020–2021 (4)
High school or more (%)	–1.49 (0.92)	0.41 (0.71)	0.77 (0.77)	–0.72 (0.95)
Some college or more (%)	0.41 (1.57)	3.42*** (1.28)	2.37** (1.19)	0.67 (1.45)
Bachelor's or more (%)	2.41 (1.48)	5.77*** (1.41)	3.07*** (1.11)	0.68 (1.24)
Master's or more (%)	0.81 (0.69)	2.27*** (0.72)	1.38** (0.59)	0.35 (0.60)
First-time four-year students (% of 18–24)	–1.27*** (0.39)	–1.10** (0.47)	–0.82 (0.56)	0.03 (0.79)
College graduates (% of employment)	3.22** (1.41)	7.65*** (1.32)	4.23*** (0.96)	1.30 (0.98)
STEM occupations (% of employment)	1.00** (0.46)	1.86*** (0.40)	0.90*** (0.29)	0.68*** (0.24)
Log median household income	0.12*** (0.04)	0.18*** (0.03)	0.09*** (0.03)	0.03 (0.03)
Poverty rate (%)	–1.48*** (0.55)	–2.29*** (0.54)	–1.18* (0.65)	0.06 (0.80)
Unemployment rate (%)	0.04 (0.26)	0.41 (0.33)	0.36 (0.29)	0.29 (0.33)
Log population	1.12*** (0.22)	1.71*** (0.17)	1.14*** (0.12)	0.49*** (0.15)
Pre-treatment years	2005–2008	2005–2012	2005–2015	2005–2018
Commuting zones in the group	25	29	39	53
Comparison commuting zones	163	134	95	42

Notes: Each cell reports the difference in the average level of the variable during the pre-treatment years between the commuting zones in the group and the comparison commuting zones. Each column compares commuting zones whose first Amazon facility opened in the indicated years with commuting zones that received their first facility later, which serve as their comparison group in the difference-in-differences design, using only years before the group is treated (treatment begins the year before the first opening). The sample is the balanced 2005–2024 panel (2020 not available) of commuting zones that receive an Amazon facility; the comparison group for the 2020–2021 group consists of commuting zones first entered in 2022–2024. Standard errors, allowing for unequal variances, in parentheses. Attainment and population are from published ACS 1-year estimates for counties with at least 65,000 residents, aggregated to commuting zones; income, poverty, unemployment, and workforce measures are from IPUMS ACS microdata; first-time four-year students (local-draw colleges, as a percentage of the population aged 18–24) are from IPEDS, available from 2006.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effects of Amazon's Entry on Educational Attainment by Gender and Race

<i>Panel A. By gender</i>				
	Bachelor's+ men (1)	Bachelor's+ women (2)	Master's+ men (3)	Master's+ women (4)
Amazon Entry	0.477** (0.204)	0.475** (0.186)	0.492*** (0.117)	0.298** (0.118)
Mean Dep. Var.	28.17	27.84	10.61	10.00
Commuting zones	188	188	188	188
<i>Panel B. Bachelor's degree or more, by race and ethnicity</i>				
	White non-Hispanic (1)	Black (2)	Hispanic (3)	
Amazon Entry	0.470** (0.203)	0.789 (0.675)	1.183* (0.649)	
Mean Dep. Var.	31.33	16.70	13.20	
Commuting zones	187	53	47	

Notes: Dependent variables are the percentage-point shares of men or women (Panel A) or of adults in the specified racial or ethnic group (Panel B) aged 25 and older with at least the specified degree, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), averaged over the post-entry event years, comparing treated commuting zones to not-yet-treated controls, in the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens. In Panel B, each column excludes commuting zones in which the Census Bureau suppresses any county's estimate for the group in any year; the estimate for Asian adults, available for 11 commuting zones, and the event-study figures are in Online Appendix Section OA1.1. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Effects of Amazon's Entry on First-Time College Enrollment

	Two-Year Colleges (1)	Four-Year Colleges (2)	Public Four-Year (3)	Full-Time First-Time Four-Year (4)
<i>Panel A. Local-draw colleges</i>				
Amazon Entry	−0.044 (0.090)	0.215* (0.117)	0.248* (0.146)	0.185* (0.095)
Mean Dep. Var.	3.88	5.08	4.94	4.78
Commuting zones (treated)	415 (185)	325 (177)	281 (158)	324 (176)
<i>Panel B. All colleges</i>				
Amazon Entry	−0.067 (0.090)	0.115 (0.133)	0.174 (0.132)	0.103 (0.101)
Mean Dep. Var.	4.03	6.96	5.54	6.51
Commuting zones (treated)	422 (187)	377 (184)	314 (169)	377 (184)

Notes: Dependent variable is first-time degree- or certificate-seeking undergraduate enrollment (column 4: full-time first-time enrollment) in the specified type of institution located in the commuting zone, as a percentage of the commuting zone's population aged 18–24, 2010–2023. Enrollment is from IPEDS Fall Enrollment (one all-students record per institution-year) aggregated to 2020 commuting zones through institution counties; population is from IPUMS ACS 1-year microdata. Two-year colleges are institutions that do not award bachelor's degrees and four-year colleges are bachelor's-granting institutions, following [Charles et al. \(2018\)](#). Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008 (IPEDS residence data), measured before any commuting zone is treated. A commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant'Anna \(2021\)](#), aggregated across treatment cohorts; the estimator differences out time-invariant commuting-zone characteristics and common year shocks by comparing changes in treated commuting zones with contemporaneous changes in comparison commuting zones. Both panels use not-yet-treated commuting zones as comparisons. The sample for each column contains commuting zones with positive enrollment of the relevant type in every year; cohorts treated in 2010 have no pre-period and do not contribute. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Household Income and the Education Effects

<i>Panel A. Household income, employment, and rents</i>						
	Log median household income (1)	Log mean wage (2)	Poverty rate (3)	Unemployment rate (4)	Labor force participation (5)	Log median gross rent (6)
Amazon Entry	0.014*** (0.005)	0.002 (0.005)	0.055 (0.150)	−0.264* (0.149)	0.278** (0.128)	0.036*** (0.006)
Mean Dep. Var.	\$35,470	\$27,058	13.44	7.09	63.16	\$534
Commuting zones	204	204	204	204	204	204
<i>Panel B. By baseline poverty rate</i>						
	Poorer commuting zones	Richer commuting zones	Difference (1) – (2)			
Log median household income	−0.001 (0.007)	0.028*** (0.006)	−0.028*** (0.010)			
Bachelor's or more, adults 25+ (pp)	0.200 (0.306)	0.562** (0.220)	−0.362 (0.377)			
Commuting zones (attainment)	95	93				

Notes: Each cell is the doubly-robust staggered difference-in-differences estimate of [Callaway and Sant'Anna \(2021\)](#), averaged over the post-entry event years, comparing treated commuting zones with not-yet-treated commuting zones; a commuting zone is treated from the year before its first Amazon facility opens. Panel A: outcomes from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024; income, wages, and rents in constant 1999 dollars; columns (1), (2), and (6) in logs, columns (3)–(5) in percentage points; Mean Dep. Var. is the pre-treatment mean (geometric mean for logs). Panel B: treated commuting zones are split at the median of the baseline (2005–2009) poverty rate (12.1 percent); poorer commuting zones are above the median. Each cell is estimated within the subsample, with not-yet-treated commuting zones in the same subsample as comparisons. Household income is from IPUMS ACS microdata; the bachelor's share is from published ACS 1-year estimates for counties with at least 65,000 residents. Column (3) tests the difference assuming independent subsamples. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Full estimates are in Appendix Tables [A7](#) and [A8](#). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effects of Amazon's Entry on Household Income, Employment, and Rents, 2005–2024

	Log median household income (1)	Log mean wage (2)	Poverty rate (3)	Unemployment rate (4)	Labor force participation (5)	Log median gross rent (6)
Average	0.014*** (0.005)	0.002 (0.005)	0.055 (0.150)	−0.264* (0.149)	0.278** (0.128)	0.036*** (0.006)
Mean Dep. Var.	\$35,470	\$27,058	13.44	7.09	63.16	\$534
Commuting zones	204	204	204	204	204	204

Notes: Dependent variables are computed from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024. Median household income and median gross rent are weighted medians across households (renters for rent) in constant 1999 dollars; the mean wage is the average wage and salary income of workers with positive earnings, in constant 1999 dollars; the poverty rate is the share of persons below the federal poverty line; unemployment and labor force participation rates refer to persons aged 16 and older. Columns (1), (2), and (6) are in logs, so coefficients are approximately proportional changes; columns (3)–(5) are in percentage points. A commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant'Anna \(2021\)](#) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons. The estimates average the post-entry event-time effects. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones, in 1999 dollars for columns (1), (2), and (6) (geometric mean). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Effects of Amazon's Entry by Baseline Poverty and Income

	Poorer commuting zones (1)	Richer commuting zones (2)	Difference (1) – (2) (3)
<i>Panel A. Split by baseline poverty rate</i>			
Log median household income	–0.001 (0.007)	0.028*** (0.006)	–0.028*** (0.010)
Bachelor's or more, adults 25+ (pp)	0.200 (0.306)	0.562** (0.220)	–0.362 (0.377)
First-time four-year enrollment	–0.009 (0.188)	0.398** (0.189)	–0.406 (0.267)
Part-time share of enrollment (pp)	–1.273** (0.514)	–1.626* (0.943)	0.353 (1.074)
Commuting zones (attainment sample)	95	93	
<i>Panel B. Split by baseline median household income</i>			
Log median household income	–0.004 (0.007)	0.037*** (0.009)	–0.041*** (0.012)
Bachelor's or more, adults 25+ (pp)	–0.017 (0.304)	0.428 (0.475)	–0.445 (0.564)
First-time four-year enrollment	0.159 (0.194)	–0.067 (0.128)	0.226 (0.233)
Part-time share of enrollment (pp)	–0.572 (0.769)	–2.468** (1.078)	1.897 (1.324)
Commuting zones (attainment sample)	88	100	

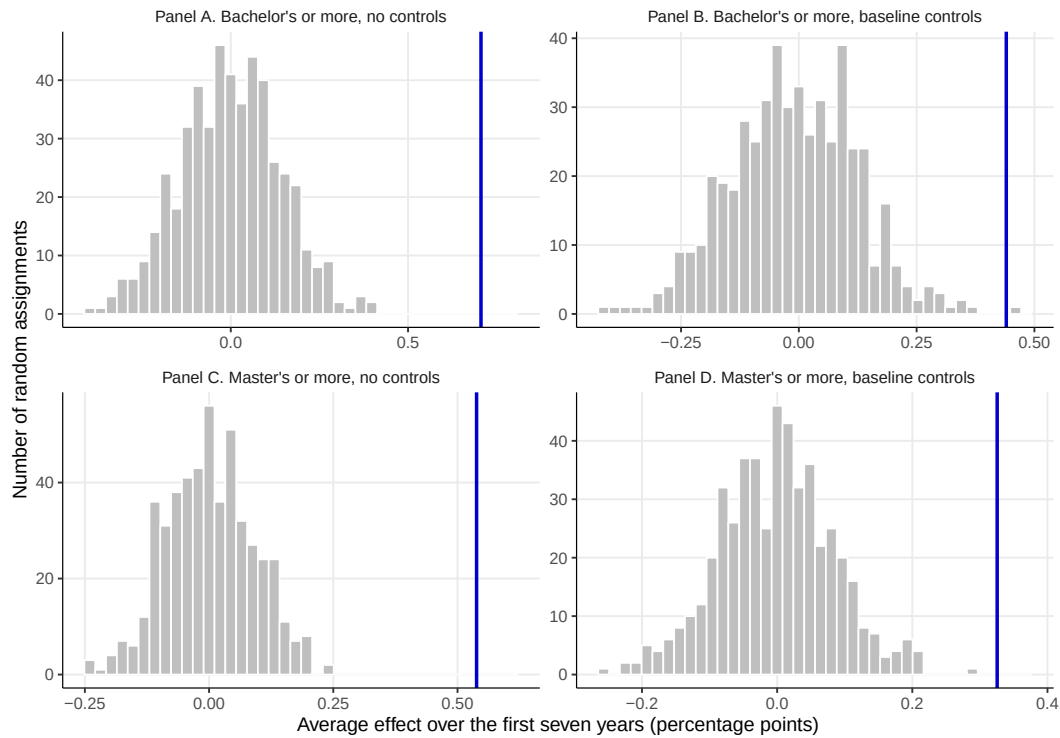
Notes: Commuting zones that receive an Amazon facility are split at the median of the baseline (2005–2009) poverty rate (Panel A; median 12.1 percent) or of baseline median household income (Panel B). Poorer commuting zones are those above the median poverty rate or below the median income; 79 percent of high-poverty commuting zones are also low-income. Each cell is a separate doubly-robust staggered difference-in-differences estimate (Callaway and Sant'Anna, 2021) within the subsample, with not-yet-treated commuting zones in the same subsample as comparisons, averaged over the post-entry event years. Household income is from IPUMS ACS microdata, 2005–2024; attainment is from published ACS 1-year estimates for counties with at least 65,000 residents, 2005–2024 (2020 not available); first-time four-year enrollment at local-draw colleges (as a percentage of the population aged 18–24) and the part-time share of enrollment are from IPEDS, 2010–2023. Column (3) tests the difference assuming independent subsamples. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Effects of Amazon's Entry on Industry Composition and Establishments

	Amazon Entry (1)	Std. error (2)	Mean before entry (3)
<i>Panel A. Industry share of employment (pp), County Business Patterns, 2005–2023</i>			
Professional, scientific & technical	0.137*	(0.073)	5.03
Information	0.068	(0.056)	2.16
Finance & insurance	0.077*	(0.040)	4.47
Management of companies	0.089*	(0.049)	1.91
Construction	0.175***	(0.057)	5.54
Retail trade	−0.121**	(0.058)	14.74
Accommodation & food	−0.194**	(0.077)	11.17
Administrative & support	−0.156**	(0.072)	5.65
Manufacturing	−0.153*	(0.080)	12.65
<i>Panel B. Establishments, Business Dynamics Statistics, 2005–2023</i>			
Professional, scientific & technical (log)	0.032***	(0.007)	1,054
Information (log)	0.021**	(0.009)	199
Finance & insurance (log)	0.015***	(0.005)	799
Manufacturing (log)	0.005	(0.007)	501

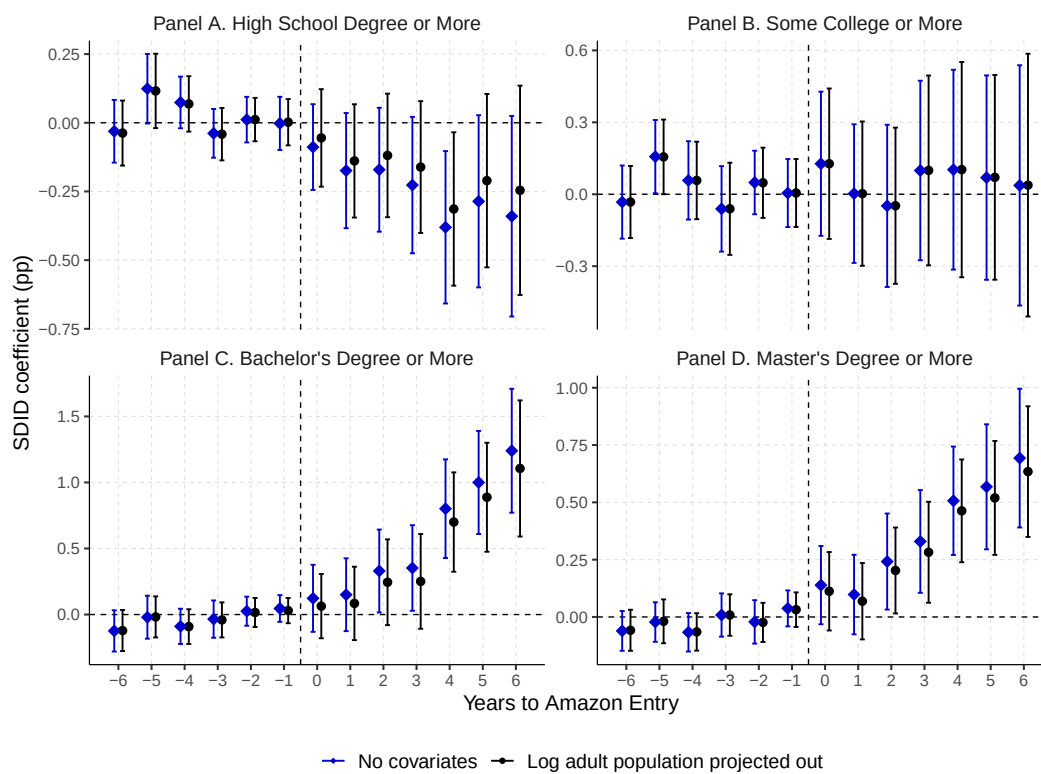
Notes: Panel A: employment in each 2-digit NAICS sector as a percentage of total employment, from County Business Patterns (mid-March employment by workplace) aggregated from counties to commuting zones; employment withheld for disclosure before 2017 is set to the midpoint of the reported size class, and commuting zones whose county composition changes over time are excluded. Panel B: log number of establishments by sector, from the Census Bureau's Business Dynamics Statistics aggregated from counties to commuting zones. A commuting zone is treated from the year before its first Amazon facility opens. Each row is a separate doubly-robust staggered difference-in-differences estimate (Callaway and Sant'Anna, 2021) on the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons; column (1) averages the event-time effects, and column (2) reports standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level. Column (3) is the pre-treatment mean in treated commuting zones (for log outcomes, the geometric mean count). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Randomization Inference: Actual Estimates versus Random Assignments of Amazon's Entry



Notes: Each histogram shows the distribution of the average post-entry effect across 500 random reassignments of treatment years among commuting zones that receive an Amazon facility, holding fixed the number of commuting zones treated in each year. The vertical line marks the actual estimate. Estimates are the doubly-robust staggered difference-in-differences estimates of [Callaway and Sant'Anna \(2021\)](#) with not-yet-treated commuting zones as comparisons; Panels B and D condition on log population in 2005–2009 and the change in log population between 2005–2006 and 2008–2009. Outcomes are the percentages of adults aged 25 and older with at least a bachelor's or a master's degree, from IPUMS ACS microdata, 2005–2024.

Figure A2: Synthetic Difference-in-Differences Estimates: Effects of Amazon’s Entry on Educational Attainment, 2005–2024



Notes: Each panel plots staggered synthetic difference-in-differences event-study estimates (Arkhangelsky et al., 2021) for the percentage of adults aged 25 and older with at least the specified level of schooling, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates at negative event times are placebo effects for the six years before treatment. Blue diamonds: no covariates; black circles: log adult population projected out. Comparison units are commuting zones that never receive an Amazon facility, reweighted to match the pre-treatment path of treated commuting zones. Bars show 95% confidence intervals from 200 bootstrap replications. The sample contains commuting zones observed in every year. Treatment begins the year before Amazon’s first facility opens.